

Graphs of functions - Set 1

TMUA practice questions by topic. Specification MM8.1, MM8.5-MM8.7, M4.15. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 How many times does the graph of $y = x^3 - x^2 - 6x$ cross or touch the x -axis?
- A** 3
B 1
C 0
D 2
E 4
- 2 The graphs of $y = x^2 - 2x$ and $y = k$ meet at exactly two points. Find the complete set of values of k .
- A** $k > -1$
B $k \geq -1$
C $k > 0$
D $k = -1$
E $k < -1$
F all real k
- 3 Which of the following could be the equation of a cubic curve that touches the x -axis at $x = 2$ and crosses it at $x = -1$, and has a negative y -intercept?
- A** $y = -(x - 2)^2(x + 1)$
B $y = (x - 2)(x + 1)^2$
C $y = (x + 2)^2(x - 1)$
D $y = -(x + 2)^2(x - 1)$
E $y = (x - 2)^2(x + 1)$

- 4 Which of the following statements about the graph of $y = |2x - 4|$ is/are true?
- I It meets the x -axis at exactly one point.
 - II It is symmetric about the line $x = 2$.
 - III It lies entirely on or above the line $y = 0$.
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III
- 5 The curve $y = x^4 - 5x^2 + 4$ meets the coordinate axes at how many points in total?
- A 5
- B 3
- C 6
- D 1
- E 2
- F 4
- 6 The graph of $y = \log_2 x$ is reflected in the line $y = x$. The resulting graph has equation
- A $y = 2^x$
- B $y = \log_{\frac{1}{2}} x$
- C $y = \log_2 (-x)$
- D $y = 2^{-x}$
- E $y = x^2$
- F $y = -\log_2 x$

- 7 A polynomial of degree 5 with real coefficients has how many real roots at least, and at most?
- A at least 1, at most 5
 B exactly 5
 C at least 1, at most 4
 D at least 0, at most 4
 E at least 0, at most 5
 F at least 2, at most 5
- 8 Which of the following functions is undefined at exactly two real values of x ?
- A $f(x) = \frac{1}{(x-2)^2}$
 B $f(x) = \frac{1}{x^2}$
 C $f(x) = \sqrt{x-2}$
 D $f(x) = \frac{1}{x^2+4}$
 E $f(x) = \frac{1}{x^2-4}$
- 9 The line $y = mx$ meets the curve $y = x^3 - x$ at three distinct points. Find the complete set of values of m .
- A $m < -1$
 B $-1 < m < 0$
 C $m > 1$
 D $m > 0$
 E $m > -1$
 F all real m
- 10 The line $y = x + c$ and the curve $y = \frac{1}{x}$ have no points of intersection. Find the complete set of values of c .
- A $c < -2$ or $c > 2$
 B $-1 < c < 1$
 C $c > 2$
 D $c < 2$
 E $-2 < c < 2$
 F no such values

- 11** The graph of $y=f(x)$ has $f'(x) > 0$ for $x < 1$, $f'(1) = 0$, $f'(x) > 0$ for $1 < x < 3$, $f'(x) < 0$ for $x > 3$. Which of the following is true?
- A** f has local maxima at $x = 1$ and $x = 3$
 - B** f has a local minimum at $x = 1$ and a local maximum at $x = 3$
 - C** f has a local maximum at $x = 3$ and a stationary point of inflexion at $x = 1$
 - D** f has a local maximum at $x = 1$ and a local minimum at $x = 3$
 - E** f is increasing for all x
- 12** How many real solutions does the equation $x^3 = 3x + 1$ have?
- A** 2
 - B** 0
 - C** 1
 - D** 3
 - E** infinitely many